Spurious or Structural?
Probability Limits and a Registered Falsification Design for Cross-Asset Price Impact

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Abstract

Cross-asset return-on-flow coefficients are tempting to read as entries of a structural price-impact matrix. In a simultaneous system, that reading can fail for two different reasons: latent common factors and same-bin feedback both create off-diagonal population coefficients when the matching structural entries are zero. We derive the probability limits of ordinary least squares and of least squares with a noisy factor proxy in an N-asset system. The decomposition shows which part of the discrepancy a noisy control can remove and which part it leaves behind. We then test the formulas in a preregistered known-truth experiment with $N = 30$, $K = 3$, and $T = 10^7$. The sole frozen draw passed the maximum elementwise relative-discrepancy threshold of 10^{-3}: the uncontrolled and proxy-controlled discrepancies were 5.6395×10^{-4} and 5.1237×10^{-4}, and all 1,800 targets fell inside named 95% family-wise Bonferroni intervals. This validates the algebra and implementation under the stated simulation law; it does not identify market impact. The paper closes by registering the next experiment: a source-constrained premise test with an observable integrated-order-flow opponent, two oracle-flow checks, six published-protocol reconstructions, whole-date bootstrap inference, and explicit size, power, and failure rules. That experiment is specified here, not reported.

Research status. This pre-results preprint reports the completed derivation and one preregistered known-truth simulation. It does not report the G2 premise test, market-data study, training sample, holdout, or economic evaluation. No registered G2 resource benchmark, validation panel, or research draw has run, and no external market data has been accessed.

Keywords: cross-impact; order-flow imbalance; latent factors; simultaneity; measurement error; preregistration; bootstrap inference.

1 Introduction

Suppose the return of asset i covaries with the order flow of asset j. The association may reflect a direct causal channel from j's flow to i's price. It may instead reflect a market-wide order-flow component, a common information shock, contemporaneous price chasing, measurement error, or some combination of these mechanisms. A multivariate regression will assign a coefficient either way. The scientific problem is therefore not whether a cross coefficient can be estimated, but which structural object that coefficient targets.

This distinction matters for both research and practice. Structural impact matrices enter execution-cost models, liquidity stress tests, and manipulation constraints. Reduced-form predictive coefficients can still be useful, but using them as structural primitives can misprice basket execution and convert commonality into an apparent transmission network. The theoretical market impact literature provides strong reasons to care about the form of cross-impact [1, 9]; the empirical literature shows both substantial commonality in returns and order flow and instability in cross coefficients [5, 7].

We study the boundary between these interpretations. The paper is organized around a simple principle: every claim must survive a known-truth or falsification gate before it can enter the empirical conclusion. Figure 1 summarizes the logic.

The first contribution is an exact probability-limit decomposition for a simultaneous multi-asset return-flow system. Ordinary least squares targets the structural impact matrix plus two terms: latent-factor confounding and same-bin simultaneity. A noisy factor control replaces factor variance with the variance remaining after linear projection on the proxy, but it does not in general eliminate either total bias or simultaneity.

The second contribution is a verified known-truth recovery exercise. The derivation was frozen before simulation code and random-number access. A single registered 10^7-observation draw at the intended $N = 30$ dimension recovered both population matrices under a strict no-floor discrepancy statistic. This is evidence about the derivation and numerical implementation, not about market structure.

The third contribution is a registered premise test that gives the confounding hypothesis a severe pre-empirical check. It asks whether confounding alone can create economically material off-diagonal error in a transparent one-

Structural system ⟶ Regression target ⟶ Known-truth recovery ⟶ Premise test ⟶ Market identification ⟶ Sealed holdout

Figure 1: Claim flow and current evidence boundary. The diagram is a research control map, not an empirical result. The derivation and known-truth recovery are complete; the premise test is registered but unrun, and all later claims remain locked.

Table 1: Research status at the manuscript freeze.

Gate	Status	Licensed statement
G0	Passed	Reproducible software and compute-plan skeleton only.
G1	Passed	Derived regression targets recover in the frozen known-truth simulation.
G2	Open	Scientific design registered; no registered benchmark, validation, or research result.
G3–G7	Locked	No market-data, identification, training, holdout, or economic claim.
G8	Locked	Final-results paper remains locked; this preprint adds no downstream result.

factor model constrained by published one-minute commonality summaries. The observable estimator receives integrated top-ten order-flow imbalance (OFI), the correct factor direction, and an independent 95%-reliable proxy. Two oracle-flow estimators remove measurement error and high-dimensional inversion as alternative explanations. A reconstruction of the published integrated cross-impact equation supplies an additional veto. The test does not infer the real market's structural tuple from aggregate source statistics.

Table 1 makes the present manuscript boundary explicit. Only the first two rows contain completed evidence.

2 Relation to the Literature

Linear impact originates as a theoretical and empirical language for how order flow moves prices [1, 2]. At the limit order book, best-level OFI compresses additions, cancellations, and executions into a signed liquidity-pressure statistic; short-interval price changes have been shown to relate strongly to this quantity [3]. Multi-level OFI extends the statistic deeper into the book [4]. Cont, Cucuringu, and Zhang [8] use principal components to integrate levels and compare own-asset and cross-asset regressions in a one-minute forecasting protocol.

Cross-sectional commonality complicates structural interpretation. Hasbrouck and Seppi [5] document common factors in prices and order flows. Benzaquen et al. [6] show how multivariate propagator models can encode cross-asset covariance and sector structure. Capponi and Cont [7] argue that a common OFI component can explain much of apparent contemporaneous cross-impact. These papers motivate the central ambiguity here: the same reduced-form covariance can be consistent with materially different structural stories.

Our objective differs from ranking predictive models. We ask whether the coefficient map produced by a specified regression equals a structural impact matrix under explicit simultaneous equations. Factor methods can summarize a large predictor panel [10], and generated-factor inference has a mature asymptotic theory [11]. Yet a predictive factor is not automatically a valid causal control, and one noisy proxy does not generically identify an unobserved confounder [12]. The theorem below makes that limitation algebraically visible in this setting.

High-dimensional cross-flow specifications require regularization. LASSO has a well-known prediction and selection role [13], including under structured time-series assumptions [14, 15]. Our registered reconstruction uses LASSO as a frozen algorithmic component, not as a structural-identification theorem. Penalty grids, fold chronology, scaling, factor penalization, solver tolerances, and tie rules are fixed before any registered output is observed.

Broad model searches can report noise as a winner [16, 17, 18]. This design limits that risk through preregistration and an intersection-union decision rule. The whole-date bootstrap preserves all within-date bins and assets; its validity is a gate condition, not something assumed from resampling alone.

3 Simultaneous System and Regression Targets

Let returns r_t, flows q_t, and idiosyncratic shocks u_t, v_t lie in $\mathbb{R}^N$, with latent factors $f_t \in \mathbb{R}^K$. Consider

$$r_t = \Lambda q_t + \Gamma f_t + u_t, \tag{1}$$
$$q_t = B r_t + \Delta_f f_t + v_t. \tag{2}$$

Here Λ is the structural contemporaneous impact matrix and B is same-bin return-to-flow feedback. All variables are centered. Assume finite second moments and a law of large numbers; mutual zero contemporaneous cross-covariances among f_t, u_t, v_t, and proxy noise; and nonsingularity of $I_N - B\Lambda$ and the relevant regressor covariance matrices. These are conditions for the stated probability limits, not empirical facts about an exchange.

Define

$$L = I_N - B\Lambda, \qquad H = L^{-1}, \qquad D = B\Gamma + \Delta_f, \tag{3}$$
$$P = HD, \qquad U = HB, \qquad V = H. \tag{4}$$

Substitution gives the flow reduced form

$$q_t = Pf_t + Uu_t + Vv_t. \tag{5}$$

Let Σ_f, Σ_u, and Σ_v denote the corresponding shock covariances. Then

$$\Sigma_{qq} = P\Sigma_f P^\top + U\Sigma_u U^\top + V\Sigma_v V^\top. \tag{6}$$

Theorem 1 (Pseudo-true cross-impact matrices)**.** *Under the conditions above, the population coefficient in the regression of r_t on q_t is*

$$\operatorname{plim} \widehat{\Lambda}_{\text{OLS}} = \Lambda + \Gamma\Sigma_f P^\top \Sigma_{qq}^{-1} + \Sigma_u U^\top \Sigma_{qq}^{-1}. \tag{7}$$

If the regression additionally controls for $h_t = f_t + \epsilon_t$, where ϵ_t is mutually uncorrelated with the system shocks, define

$$R_f = \Sigma_f - \Sigma_f(\Sigma_f + \Sigma_\epsilon)^{-1}\Sigma_f \tag{8}$$

and

$$Q_h = PR_f P^\top + U\Sigma_u U^\top + V\Sigma_v V^\top. \tag{9}$$

The coefficient on q_t after controlling for h_t is

$$\operatorname{plim} \widehat{\Lambda}_{q|h} = \Lambda + \Gamma R_f P^\top Q_h^{-1} + \Sigma_u U^\top Q_h^{-1}. \tag{10}$$

Equation (7) has two distinct sources of discrepancy from Λ. The first is latent-factor confounding. The second arises because the return shock u_t enters flow through same-bin feedback B. In Eq. (10), the proxy replaces Σ_f by the residual factor variance R_f after linear projection. It leaves the feedback channel available and changes the inverse covariance that weights both terms.

Corollary 1 (Perfect and useless controls)**.** *If $\Sigma_\epsilon = 0$, then $R_f = 0$ and factor confounding disappears, but the simultaneity term remains unless its numerator is zero. If proxy noise diverges in every direction, then $R_f \to \Sigma_f$ and the controlled coefficient converges to the uncontrolled coefficient.*

Corollary 2 (Spurious cross coefficients)**.** *Setting a structural entry $\Lambda_{ij} = 0$ does not imply that the population regression coefficient at (i, j) is zero. Either bias term in Eq. (7) can be off-diagonal, including when all structural off-diagonals are zero.*

Corollary 3 (No-feedback benchmark)**.** *If $B = 0$, the simultaneity term vanishes. A perfect factor proxy is then sufficient for recovery of Λ under the stated covariance restrictions.*

Proposition 1 (Proxy precision is not elementwise bias monotonicity)**.** *Suppose Σ_f and two proxy-noise covariance matrices are positive definite, with $\Sigma_{\epsilon,1} \preceq \Sigma_{\epsilon,2}$. The associated residual factor covariances satisfy $R_{f,1} \preceq R_{f,2}$. Nevertheless, the absolute value of each entry of $\operatorname{plim} \widehat{\Lambda}_{q|h} - \Lambda$ need not be ordered because Q_h^{-1} changes with proxy precision and reweights both bias components.*

The proposition is a warning against scalar attenuation intuition. A better proxy removes residual factor variation in positive-semidefinite order, but a matrix coefficient is a ratio of covariance objects. Structural interpretation requires the full map.

Table 2: Completed G1 known-truth verification.

Quantity	Verified value
Assets / factors / observations	$30/3/10^7$
Frozen master seed	2026071501
Reported coefficients	1,800
OLS max relative discrepancy	5.6394671×10^{-4}
Proxy-control max discrepancy	5.1237142×10^{-4}
Gate threshold	1.0×10^{-3}
Targets inside simultaneous intervals	1,800 / 1,800
Interval method	Student-t, Bonferroni 95% FWER

4 Known-Truth Verification

The probability limits were evaluated along two algebraically independent population paths and then tested in a frozen simulation. The registered fixture used $N = 30$ assets, $K = 3$ factors, and $T = 10{,}000{,}000$ independent Gaussian observations. The matrices were dense and deliberately nontrivial, with $\rho(B\Lambda) = 0.4104$ and condition numbers approximately 348.1 for the uncontrolled flow covariance and 92.4 for the proxy-residualized flow covariance.

The gate statistic was the maximum elementwise relative discrepancy across the uncontrolled and controlled coefficient matrices,

$$D_{\text{gate}} = \max_{m \in \{\text{OLS,proxy}\}} \max_{i,j} \frac{|\widehat{A}_{m,ij} - A^*_{m,ij}|}{|A^*_{m,ij}|}, \tag{11}$$

with no denominator floor, element exclusion, averaging substitute, or seed retry. Passage required $D_{\text{gate}} < 10^{-3}$.

The sole frozen draw passed. Table 2 reports the complete gate-level summary. All 1,800 target coefficients were inside the named 95% family-wise classical homoskedastic Student-t Bonferroni intervals. The Gaussian law makes that interval method appropriate for this controlled fixture; the method is not transported to dependent market data.

This result establishes three narrow facts. First, the symbolic expressions and the primitive covariance implementation agree. Second, row-stacked software returns the transpose expected from the column-vector structural notation, and that orientation is handled correctly. Third, the streamed sufficient-statistic and publication paths recover the registered target at the intended dimension.

It does not establish identification, empirical fit, causal timing, interval coverage under market dependence, or economic usefulness. The fixture's target coefficients are large and positive, so it does not resolve the near-zero, sign-sensitive problem that motivates the next gate. One draw also cannot estimate repeated-sampling coverage. The completed experiment is therefore a derivation and implementation validation, not a market result.

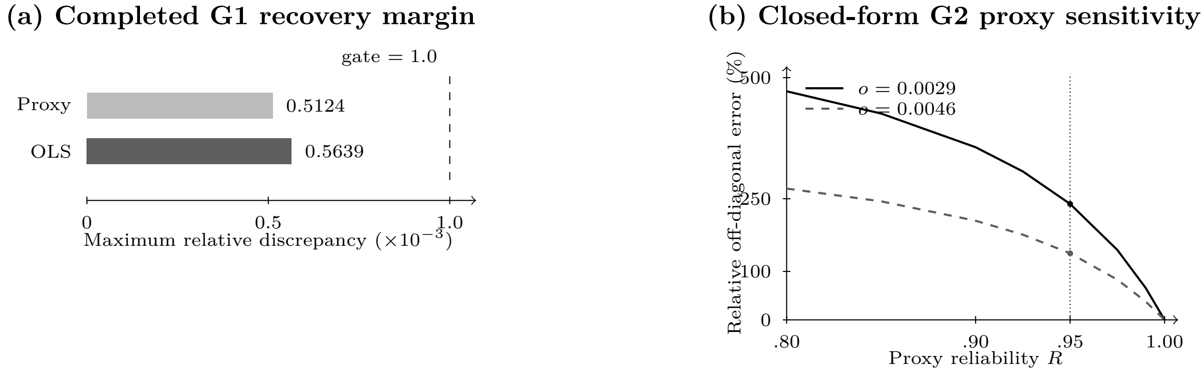

Figure 2: Two different kinds of quantitative evidence. Panel (a) reports the completed G1 gate statistic; both bars lie below the preregistered threshold. Panel (b) evaluates Eq. (19) over proxy reliability for the two registered structural endpoints. The dotted line marks $R = 0.95$. Panel (b) is an analytic pre-run prediction, not a stochastic G2 result or a market estimate.

5 Registered Premise Test: Design, Not Result

The next gate asks whether the motivating confounding mechanism survives a severe conditional test before empirical data is opened. The question is existential rather than identificational:

> Can confounding alone be materially large in a transparent model whose observable flow commonality and factor alignment match opened primary-source summaries, even when the estimator receives a favorable factor proxy?

The test removes same-bin feedback by setting $B = 0$. It therefore cannot pass because of price chasing. The primary estimator observes noisy integrated OFI, the correct factor direction, and an independent 95%-reliable proxy. Oracle estimators observe true flow and include a symmetry-aware low-dimensional fit. A positive result would be conditional evidence for one source-compatible model class, not an estimate of the market's Λ.

5.1 Permutation-invariant source constraint

Let $N = 30$ and $m = \mathbf{1}_N/\sqrt{N}$. For leading flow- and return-correlation shares s_q and s_r, define

$$q_1 = N s_q, \qquad q_0 = \frac{N - q_1}{N - 1}, \tag{12}$$

$$r_1 = N s_r, \qquad r_0 = \frac{N - r_1}{N - 1}. \tag{13}$$

The registered maximum-entropy one-spike convention is

$$\Sigma_{qq} = q_0 I + (q_1 - q_0) m m^\top, \tag{14}$$

$$\Sigma_{rr} = r_0 I + (r_1 - r_0) m m^\top. \tag{15}$$

At the one-minute reference point, $(s_q, s_r, \rho) = (0.2827, 0.32, 0.8726)$. The implied pairwise flow correlation is 0.25797, reproducing the reported 0.26 to its published precision [7]. The isotropic residual spectrum is a declared modeling choice because the source summary does not identify its orientation or lower eigenvalues.

Set diagonal structural sensitivity $d = 0.29$, off-diagonal sensitivity $o \in [0.0029, 0.0046]$, and

$$\Lambda = (d - o) I + N o\, m m^\top. \tag{16}$$

These values are sensitivities, not one-minute structural estimates. The 1% cross-to-own lower ratio is an economic-materiality floor; the upper ratio is motivated by a normalized homogeneous propagator estimate [6].

Let

$$q = h_q m f + v, \qquad \mathrm{Var}(v) = q_0 I, \tag{17}$$

$$r = \Lambda q + \gamma m f + u, \tag{18}$$

where $\mathrm{Var}(f) = 1$ and the common return-flow covariance fixes γ. The observable control is $z = f + e$ with reliability $R = \mathrm{Var}(f)/\mathrm{Var}(z)$. At the registered $R = 0.95$, the estimator is not told the proxy-error variance and makes no reliability correction.

Proposition 2 (Off-diagonal proxy-control target)**.** *In the homogeneous no-feedback model above, the population off-diagonal in the regression of r on (q, z) is*

$$\widehat{o}_R = o + \frac{(1 - R)(c_1 - \lambda_1 q_1)}{N\{q_0 + (1 - R)(q_1 - q_0)\}}, \tag{19}$$

where $\lambda_1 = d + (N-1)o$ and $c_1 = \rho\sqrt{r_1 q_1}$. If $c_1 - \lambda_1 q_1 > 0$, the bias is positive for every $R < 1$, vanishes at $R = 1$, and the relative error decreases strictly with reliability. At the registered reference point, it also decreases with o, so the upper endpoint is least favorable over the structural interval.

Equation (19) produces sealed pre-run population predictions. They are not stochastic G2 results and are not estimates of a real-market decomposition. Their role is to ensure that the finite-sample experiment cannot move the target after observing output.

5.2 Strongest-opponent estimators

Three smooth estimators must all pass at all 17 structural grid points:

1. `ORACLE_Q_PROXY95_CONDITION_RIDGE` receives all true flows and the noisy factor proxy.
2. `ORACLE_Q_PROXY95_HOMOGENEOUS_OLS` is additionally told the true homogeneous symmetry and estimates only own-flow, sum-of-other-flow, and factor slopes.
3. `INTEGRATED_OFI_PROXY95_CONDITION_RIDGE` observes only date-estimated, L1-normalized top-ten principal-component scores and the proxy.

The first oracle removes OFI measurement error. The second removes both that error and the 30-dimensional covariance inversion. The third is the binding observable opponent. No post-draw selector can choose among them.

The condition-ridge estimator partials out the unpenalized proxy from globally centered sufficient statistics. If S is the residualized flow covariance, with extreme eigenvalues $s_{\max}$ and $s_{\min}$, the registered penalty is

$$\lambda_{\text{cond}} = \max\left\{0, \frac{s_{\max} - 10^4 s_{\min}}{10^4 - 1}\right\}, \tag{20}$$

$$\lambda_{\text{floor}} = 10^{-6}\frac{\operatorname{tr}(S)}{N}, \tag{21}$$

$$\lambda_{\text{ridge}} = \max\{\lambda_{\text{cond}}, \lambda_{\text{floor}}\}. \tag{22}$$

The positive floor is fixed before the draw. Its population target is sealed, so shrinkage cannot be blamed or tuned after the result.

5.3 Published-protocol reconstruction

Cont, Cucuringu, and Zhang [8] publish integrated multilevel and cross-sectional-factor specifications separately. They do not publish one estimator combining integrated top-ten OFI with a cross-sectional factor control. We therefore do not attribute the strengthened observable opponent above to that paper. Instead, the six equations in Table 3 are reconstructed separately under frozen numerical choices. The label is *protocol reconstruction*, not byte-exact author code.

The CI_I coefficient at the predeclared $(0,1)$ position is the binding published direct-flow veto because it is the only reconstructed equation with both integrated top-ten OFI and explicit off-diagonal coefficients. The other five specifications remain mandatory diagnostics but cannot rescue a failed veto.

LASSO uses five contiguous six-bin validation folds and a common 40-point ratio grid. Every PCA, mean, scale, and residualization map is fit inside the corresponding training boundary. The selected ratio minimizes pooled validation sum of squared errors under a frozen lowest-index tie rule. Fold paths warm-start in descending penalty order, while every outer refit begins at zero. The factor is unpenalized where present. Original-unit coefficients and training-defined transforms score the next block. These details matter because the source paper does not disclose a unique scaling, fold, grid, tie, or solver implementation.

For factor-residual models, a residual coefficient matrix is not relabeled as unrestricted Λ. If W is the cross-sectional loading and $P_\perp = I - WW^\top$, the purged operators are compared with $\Lambda P_\perp$. Full response-equivalent maps are reported separately. Products are formed before block and date averaging.

5.4 Decision rule and validation license

For focal estimate $\widehat{\Lambda}_{01}$, truth o, and standard error from 499 whole-date bootstrap replicates, passage requires

$$|\widehat{\Lambda}_{01} - o| - 0.50|o| > 3\,SE_{\text{boot}}. \tag{23}$$

The materiality margin beyond 50% error, not error relative to zero, must clear three standard errors. The same date weights are shared across candidates and the CI_I veto. Reported intervals are a bootstrap-standard-error normal interval and a basic whole-date bootstrap interval with named quantile rules.

Before the research draw can exist, 100 independent validation superpanels must license the exact final procedure. A null-grid superpanel evaluates 459 strict events and records whether any passes. The one-sided 95% Clopper-Pearson upper bound on that family-union probability must be at most 5%, which at $n = 100$ permits at most one union success. A power superpanel evaluates all 51 actual alternative components and records whether every component passes. The one-sided 95% Wilson lower bound on that intersection probability must be at least 80%, which requires at least 87 successes. This is a finite registered grid, not a continuum-uniform size proof.

The expensive CI_I branch receives a separate full-dimension, full-time recovery panel with confounding removed but the same collinear flow law. It must recover 30 diagonal truths and the focal off-diagonal truth inside 31-component Bonferroni bootstrap-normal intervals, with every point error strictly below 50%. One recovery panel is a no-strawman check, not a Monte Carlo size or power statement.

5.5 Outcome taxonomy fixed before results

The interpretation of each possible branch is fixed in advance:

- If all smooth candidates and CI_I pass their licensed scopes, G2 supports material confounding only in the registered source-matched model class.
- If any binding candidate fails, G2 does not pass. A nonbinding diagnostic cannot rescue it.
- If size, power, numerical, provenance, or resource admission fails, the research draw remains sealed. This is a design failure, not a market null.

Table 3: Frozen six-specification reconstruction. Each date has eleven 30-minute blocks; blocks 0–9 train and blocks 1–10 score. Every row fits 30 separate response equations.

ID	Feature map	Fit	Unpenalized	Penalized
PI_1	Own best-level OFI	Per-response OLS	Intercept, own flow	None
PI_I	Own integrated top-ten OFI	Per-response OLS	Intercept, own flow	None
CI_1	All best-level OFIs	Per-response LASSO	Intercept	All 30 flows
CI_I	All integrated top-ten OFIs	Per-response LASSO	Intercept	All 30 flows
PI_{CC}	Cross-sectional PC1 and own residual	Per-response OLS	All columns	None
CI_{CC}	Cross-sectional PC1 and all residuals	Per-response LASSO	Intercept, PC1	All 30 residuals

Table 4: Deterministic software verification on 6 August 2026.

Check	Fresh result
Ruff lint	Pass
Ruff format	26 files checked
Strict mypy	Pass, 26 source files
Pytest	306 passed in 36.99 s
Deterministic G0 demo	64 rows; expected hashes reproduced
Committed-result drift	Pass; no drift

- If the sole research draw misses after adequate validation, the finite demonstration fails. It does not logically prove the absence of population bias.
- A premise-killing null requires a separately preregistered sharp upper bound over the source-compatible class. A single negative fixture is insufficient.

6 Reproducibility and Research Controls

The project makes reproducibility a condition for every stochastic claim. A run is licensed only after its derivation, quantitative prediction, configuration bytes, input-source hashes, random-number address map, interval method, failure rules, and compute budget are frozen. Registered streams are separated by phase and scenario. Resume boundaries store sufficient statistics and provenance rather than mutable generator state. Final artifacts are published success-last and are accepted only when their markers bind the payload hashes.

The completed G1 experiment used 100 immutable shards of 100,000 observations. Each component stream was addressed by master seed, shard index, and component identifier. Aggregation reread every shard in index order and merged means and centered scatters with the between-shard correction. Replaying the completed run left the result artifacts byte-identical.

Table 4 records the fresh deterministic repository gate used for this version. These checks support software and artifact consistency; they are not additional scientific trials and do not license a G2 stream.

G2 is more demanding because the final statistic combines date-level dependence, measured-flow construction, high-dimensional fits, and bootstrap aggregation. The registered implementation is consequently required to:

- preserve whole dates as the dependence unit;
- refit all training-defined transforms inside each fold and outer block;
- use one shared set of date weights across all binding candidates;
- fail rather than drop a nonfinite, singular, weak-PCA, or nonconvergent cell;
- checkpoint long work in bounded, hash-validated units; and
- remain below the stated memory, disk, download, and wall-clock limits.

These controls do not make an estimator scientifically valid. They make it possible to tell which estimator and random variable produced a future claim.

7 Path to the Empirical Study

The empirical phase is intentionally absent from this draft. Before any market claim, the remaining gates require:

1. a byte-level audit of exactly authorized external files, timestamps, coverage gaps, and point-in-time availability;
2. a simulation identification proof based on the true-parameter Jacobian and its full singular spectrum;
3. explicit numerical demonstrations of at least two failure modes of the naive estimator;
4. train-only estimation with all attempted specifications logged and the holdout sealed;
5. theoretical admissibility, dependence-robust inference, manipulation-cost accounting, and liquidation-regret evaluation; and
6. a single holdout opening with no post-open tuning.

No trading profitability, transaction-cost saving, patentability, or deployable execution rule follows from the present evidence. Those questions become scientifically meaningful only after identification and out-of-sample gates survive.

8 Limitations

The completed result is deliberately narrow. The G1 fixture is Gaussian, known-truth, and large-sample. Its zero contemporaneous shock cross-covariances are substantive assumptions. Its dense positive targets do not reproduce the small, sign-sensitive coefficients central to the empirical debate. The Bonferroni interval inclusion is one realization, not an estimated coverage probability.

The G2 design also has sharp limits even if it eventually passes. Its one-spike residual spectrum is a declared maximum-entropy convention, not an identified feature of a market. The 95% proxy reliability is favorable to the opponent but is not empirically estimated. The structural diagonal and cross-to-own interval are sensitivity inputs assembled from nonidentical studies and units. The AR(1) dependence parameter is a hostile stress value, not a calibration. The null-grid validation is finite. The conditional model class cannot stand in for every source-compatible latent decomposition.

Finally, predictive performance and structural identification remain distinct. An estimator can forecast well while targeting the wrong impact matrix, and a structurally interpretable estimator can be economically useless after costs. Both questions must be evaluated, but neither can substitute for the other.

9 Conclusion

Off-diagonal return-on-flow coefficients do not identify structural cross-impact by notation alone. In a simultaneous multi-asset system, their probability limits contain separate latent-factor and feedback terms. A noisy factor control reduces the factor variation left after projection; it does not generally recover the structural matrix. A preregistered $N = 30$, $K = 3$, $T = 10^7$ known-truth experiment recovered the derived matrices under its frozen criterion, establishing a sound algebraic and numerical starting point.

The economically important claim remains unresolved. The next step is the registered premise test, which can reject the research program before market data is used. It specifies observable and oracle opponents, a published-protocol veto, named bootstrap inference, and explicit failure branches. Until those branches and the later empirical gates close, the conclusion is methodological: apparent cross-impact must be separated from structural impact by derivation, known-truth recovery, and adversarial falsification.

Declarations

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A Proof of Theorem 1

From Eqs. (1)–(2),

$$(I - B\Lambda)q_t = (B\Gamma + \Delta_f)f_t + Bu_t + v_t, \tag{24}$$

which yields Eq. (5). Mutual zero cross-covariances imply

$$\mathrm{Cov}(f_t, q_t) = \Sigma_f P^\top, \tag{25}$$
$$\mathrm{Cov}(u_t, q_t) = \Sigma_u U^\top. \tag{26}$$

Using $r_t = \Lambda q_t + \Gamma f_t + u_t$,

$$\Sigma_{rq} = \Lambda\Sigma_{qq} + \Gamma\Sigma_f P^\top + \Sigma_u U^\top. \tag{27}$$

Right multiplication by Σ_{qq}^{-1} proves Eq. (7).

For the proxy-controlled regression, the linear-projection residual of f_t after controlling for $h_t = f_t + \epsilon_t$ is

$$f_t^\perp = f_t - \Sigma_f(\Sigma_f + \Sigma_\epsilon)^{-1}h_t, \tag{28}$$

with variance R_f. Frisch-Waugh-Lovell residualization gives

$$q_t^\perp = Pf_t^\perp + Uu_t + Vv_t, \tag{29}$$
$$r_t^\perp = \Lambda q_t^\perp + \Gamma f_t^\perp + u_t. \tag{30}$$

Thus $\mathrm{Var}(q_t^\perp) = Q_h$ and

$$\mathrm{Cov}(r_t^\perp, q_t^\perp) = \Lambda Q_h + \Gamma R_f P^\top + \Sigma_u U^\top. \tag{31}$$

Right multiplication by Q_h^{-1} proves Eq. (10). □

B Equivalent Primitive Forms

The reduced-form matrices can be eliminated. Define

$$\Omega_q = D\Sigma_f D^\top + B\Sigma_u B^\top + \Sigma_v, \tag{32}$$
$$\Omega_{q|h} = DR_f D^\top + B\Sigma_u B^\top + \Sigma_v. \tag{33}$$

Because $\Sigma_{qq} = H\Omega_q H^\top$ and $Q_h = H\Omega_{q|h}H^\top$, Theorem 1 is equivalently

$$\operatorname{plim}\widehat{\Lambda}_{\mathrm{OLS}} = \Lambda + (\Gamma\Sigma_f D^\top + \Sigma_u B^\top)\Omega_q^{-1}(I - B\Lambda), \tag{34}$$
$$\operatorname{plim}\widehat{\Lambda}_{q|h} = \Lambda + (\Gamma R_f D^\top + \Sigma_u B^\top)\Omega_{q|h}^{-1}(I - B\Lambda). \tag{35}$$

These forms show that the decomposition is fully determined by structural primitives under the stated zero-cross-covariance assumptions.

C Frozen Numerical Rules for the Six-Specification Path

This appendix records the numerical choices that make the protocol reconstruction reproducible. It does not report a reconstruction result.

For each penalized response and training sample, the unpenalized intercept and, where applicable, factor are partialled out. Penalized columns are divided by their pre-residualization root-mean-square scale. The largest penalty is

$$\lambda_{\max} = \max_j \left| x_j^\top y/n \right|, \tag{36}$$

and the path uses the 40 frozen ratios corresponding mathematically to $10^{-4k/39}$, $k = 0, \ldots, 39$. Runtime values are read from the literal sealed vector rather than regenerated from the formula.

Coordinate descent updates, in ascending coordinate order,

$$\rho_j = x_j^\top (y - X\beta + x_j\beta_j)/n, \tag{37}$$

$$\beta_j = \frac{\operatorname{sign}(\rho_j)\max\{|\rho_j| - \lambda, 0\}}{x_j^\top x_j/n}. \tag{38}$$

Convergence requires both a scaled maximum coefficient update at most 10^{-10} and maximum KKT violation at most 10^{-9}, with 10,000 full sweeps as a hard failure. A selected outer model starts from zero even though fold paths use validated warm starts. OLS uses a full-rank least-squares solve with $rcond = \epsilon_{\text{mach}} \max(\text{shape})$.

Within-asset and cross-sectional principal components use training-column centering, covariance divisor n, a symmetric eigenvalue decomposition, the largest eigenpair, and a deterministic sign whose largest-absolute loading is positive. The top eigengap must exceed 10^{-10} times covariance trace. Validation and next-block features apply stored training means and loadings; they never refit on scored rows.

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